JanGyan — Learn Without Limits

Category 6: Environment & Civic Responsibility

Module 5 - Energy Conservation & Renewable Energy

Module Overview

This handbook is your complete, beginner-to-advanced guide to **Energy Conservation & Renewable Energy** — how India generates and uses power, why the shift to clean energy matters, and the practical steps every household can take to save electricity and support renewables. Whether you are a student, a homeowner, a working professional, or a civic volunteer, this handbook is written so you can read it once and walk away confident and capable.

By the end of this module, you will not need to search anywhere else to understand the fundamentals of energy conservation and renewable energy in India.

Learning Objectives

After completing this module, you will be able to:

- Distinguish between renewable and non-renewable energy sources
- Explain how solar, wind, and hydropower generate electricity
- Apply practical energy-efficiency and electricity-conservation habits
- Identify energy-saving appliances and what their ratings mean
- Understand India's major renewable energy programmes and targets
- Recognise the government bodies driving India's energy transition
- Apply daily energy-saving practices at home and at work
- Know where to find credible information on India's clean energy progress

Why This Topic Matters

Energy powers nearly everything in modern life — homes, industry, transport, and digital infrastructure — yet how that energy is generated has enormous consequences for air quality, climate change, and long-term energy security. India is in the middle of one of the fastest clean energy transitions in the world.

> ***Reality Check:*** *India's non-fossil fuel power capacity crossed 283 GW by March 2026, with the country adding a record 55.3 GW of clean capacity in a single financial year — the highest ever recorded. India met over half of its total electricity demand from renewables for the first time in July 2025, and now ranks third globally in installed renewable energy capacity (Ministry of New and Renewable Energy, 2026).*

Even as generation shifts toward cleaner sources, conservation remains the fastest and cheapest way to reduce both cost and environmental impact — the energy never used is energy that never needed to be generated at all.

This is why JanGyan continues its Environment & Civic Responsibility category with this essential, practical module.

Introduction

Energy conservation means using less energy to achieve the same outcome — through efficient appliances, mindful habits, and smarter design. **Renewable energy** refers to power generated from naturally replenishing sources such as sunlight, wind, and flowing water, as opposed to finite fossil fuels like coal, oil, and natural gas.

In simple words: conservation is about using less, and renewables are about generating power more cleanly. Together, they form the two pillars of a sustainable energy future — reducing demand while cleaning up supply.

India's energy story is unusual: it must expand electricity access for a still-growing population and economy while simultaneously cutting emissions — making both conservation and renewable expansion equally urgent priorities, not competing ones.

Fundamental Concepts

Before going deeper, let us understand a few foundational ideas that the entire subject of energy is built upon.

Renewable vs Non-Renewable Energy

The distinction between these two categories underpins the global shift toward sustainable energy systems.

Renewable Energy	Non-Renewable Energy
Naturally replenished (solar, wind, water, biomass)	Finite, formed over millions of years (coal, oil, natural gas)
Low to zero emissions during generation	High greenhouse gas emissions
Sustainable as a long-term energy source	Depleting at current global consumption rates

Key Definitions

- **Installed Capacity** — The maximum electricity output a power source can generate, measured in megawatts (MW) or gigawatts (GW).
- **Non-Fossil Capacity** — Combined renewable and nuclear power capacity, which does not burn fossil fuels.
- **Grid** — The interconnected network that transmits electricity from generation sources to consumers.
- **Energy Efficiency** — Using less energy to perform the same task or deliver the same output.
- **Star Rating (BEE)** — India's official labelling system showing an appliance's energy efficiency, from 1 star (least efficient) to 5 stars (most efficient).

Complete Detailed Explanation

Part A — India's Renewable Energy Mix, In Depth

As of March 2026, India's non-fossil power capacity stood at over 283 GW, made up of roughly 274.68 GW of renewable energy and about 8.78 GW of nuclear power. Within renewables, solar leads by a wide margin, followed by wind, large hydro, bio energy, and small hydro.

Source	Approximate Installed Capacity (March 2026)
Solar Power	150.26 GW
Wind Power	56.09 GW
Large Hydropower	51.41 GW
Bio Energy	11.75 GW
Small Hydro Power	5.17 GW
Nuclear Power (non-fossil, not classified as renewable)	8.78 GW

Part B — The Ten Building Blocks of Module 5

This is the heart of the module. Each topic below is explained with its core meaning and its practical relevance for Indian households and communities.

1. Types of Energy

Energy exists in many forms — mechanical, thermal, chemical, electrical, and nuclear — that can be converted from one form to another to perform useful work, such as heating, lighting, and powering machinery.

2. Renewable vs Non-Renewable Energy

Renewable sources like solar, wind, and hydropower naturally replenish and produce little to no emissions during generation, while non-renewable sources like coal and oil are finite and carbon intensive — this distinction drives India's ongoing energy transition.

3. Solar Energy

Solar energy harnesses sunlight through photovoltaic panels or solar thermal systems to generate electricity or heat. It is India's fastest-growing energy source, now the largest single contributor to the country's renewable capacity.

4. Wind Energy

Wind energy uses turbines to convert the kinetic energy of moving air into electricity. India is among the world's largest wind power markets, with strong growth concentrated in states like Gujarat, Tamil Nadu, and Karnataka.

5. Hydropower

Hydropower generates electricity from the energy of flowing or falling water, typically through dams. It remains a significant, reliable contributor to India's renewable capacity, though large projects can carry ecological and social considerations.

6. Energy Efficiency

Energy efficiency means using less energy to achieve the same output — through better insulation, efficient appliances, smart lighting, and optimised industrial processes — reducing both costs and environmental impact.

7. Electricity Conservation

Conserving electricity reduces demand on the grid, lowers household bills, and cuts the emissions associated with the fossil-fuel share still in India's overall generation mix. Simple daily habits make a measurable collective difference.

8. Energy Saving Appliances

Star-rated (BEE-rated) appliances such as LED bulbs, 5-star air conditioners, and inverter technology consume significantly less electricity than conventional appliances while delivering the same performance.

9. Government Renewable Energy Programmes

Programmes such as PM Surya Ghar: Muft Bijli Yojana (rooftop solar for households) and the National Green Hydrogen Mission are accelerating India's clean energy transition while expanding citizen participation in generation.

10. Daily Energy Saving Practices

Turning conservation into a habit — switching off unused appliances, using natural light, setting efficient thermostat temperatures — is what ultimately determines whether national energy policy translates into real household savings.

Important Terminologies

Term	Simple Meaning
Installed Capacity	The maximum electricity output a power source can generate (MW/GW)
Non-Fossil Capacity	Combined renewable and nuclear power capacity
Grid	The interconnected network transmitting electricity from source to consumer
Photovoltaic (PV)	Technology that converts sunlight directly into electricity
Net Metering	A billing system that credits rooftop solar owners for excess electricity sent back to the grid
Star Rating (BEE)	India's energy-efficiency labelling system for appliances (1–5 stars)
Distributed Renewable Energy (DRE)	Small-scale renewable installations like rooftop solar, located close to where power is used
Green Hydrogen	Hydrogen fuel produced using renewable electricity, with no associated carbon emissions
NDC (Nationally Determined Contribution)	A country's climate action targets submitted under the Paris Agreement
kWh (Kilowatt-hour)	The standard unit used to measure and bill electricity consumption

Step-by-Step Understanding

How Rooftop Solar Powers a Home

1. Sunlight Capture — Photovoltaic panels on the rooftop absorb sunlight

2. Conversion — Panels convert sunlight into direct current (DC) electricity

3. Inversion — An inverter converts DC electricity into usable alternating current (AC)

4. Consumption — The home draws power directly from the solar system during the day

5. Net Metering — Any surplus electricity is sent back to the grid, and the meter credits the household accordingly

How to Reduce Your Household Energy Footprint

1. Switch to LED bulbs and BEE 5-star rated appliances

2. Switch off lights, fans, and electronics when not in use

3. Set air conditioners to an optimal temperature (24–26°C)

4. Use natural light and ventilation during the day wherever possible

5. Service appliances regularly to maintain their efficiency

6. Consider rooftop solar where feasible, especially under government subsidy schemes

Real-Life Indian Examples

(Widely known public examples, described for educational awareness.)

- **PM Surya Ghar: Muft Bijli Yojana:** Launched in February 2024 with an outlay of over ■75,000 crore, this scheme aims to install rooftop solar on one crore households by FY 2026–27, giving families the ability to generate their own electricity and earn credits through net metering.
- **India's Record Clean Energy Year:** FY 2025–26 saw India add 55.3 GW of non-fossil capacity — the highest annual increase ever recorded — driven largely by record solar additions of around 45 GW in a single year.
- **Highest-Ever Renewable Share Day:** On 29 July 2025, renewables met 51.5% of India's total electricity demand for the first time, a symbolic milestone in the country's energy transition.
- **Ahead-of-Schedule NDC Target:** India reached 50% of its cumulative installed electricity capacity from non-fossil sources in June 2025 — five years ahead of its 2030 Paris Agreement commitment.

Practical Scenarios

Scenario 1: Your household electricity bill has been rising steadily over the past few months.

Check for old, inefficient appliances, inspect for AC or refrigerator maintenance needs, and review whether lights and fans are being left on unnecessarily.

Scenario 2: You are considering installing rooftop solar but are unsure where to start.

Check your eligibility and subsidy details under PM Surya Ghar: Muft Bijli Yojana through your state electricity distribution company or the scheme's official portal.

Scenario 3: You are shopping for a new refrigerator or air conditioner.

Check the BEE star rating label — a 5-star appliance typically consumes noticeably less electricity over its lifetime than a lower-rated one, even if the upfront cost is slightly higher.

Government Rules / Guidelines

- **Ministry of New and Renewable Energy (MNRE):** The apex national body for renewable energy policy, programme implementation, and India's non-fossil capacity targets.
- **Ministry of Power:** Oversees electricity generation, transmission, distribution, and energy conservation policy nationally.
- **Bureau of Energy Efficiency (BEE):** Administers the star-rating labelling programme for appliances and drives national energy efficiency standards.
- **Central Electricity Authority (CEA):** Plans and monitors India's power sector development, including the shift toward renewable-heavy generation.

Laws & Policies

Law / Policy	What It Covers
Energy Conservation Act, 2001 (as amended 2022)	Establishes the legal framework for energy efficiency standards, labelling, and the Bureau of Energy Efficiency
Electricity Act, 2003	Governs generation, transmission, distribution, and regulation of electricity, including renewable purchase obligations
National Electricity Policy	Sets the broad framework for expanding access, reliability, and sustainability of electricity supply
PM Surya Ghar: Muft Bijli Yojana, 2024	A national scheme subsidising rooftop solar installation for one crore households by FY 2026–27
National Green Hydrogen Mission, 2023	Aims to make India a global hub for green hydrogen production and export
India's NDC under the Paris Agreement	Commits India to 500 GW of non-fossil electricity capacity by 2030

> *India achieved 50% of its cumulative installed electricity capacity from non-fossil sources in June 2025 — five years ahead of the 2030 target set under its Nationally Determined Contribution (NDC) to the Paris Agreement.*

Important Statistics

Metric	Figure
India's total non-fossil power capacity (March 2026)	Over 283 GW
India's total renewable energy capacity (March 2026)	About 274.68 GW
Non-fossil capacity added in FY 2025–26 (record year)	55.3 GW — highest ever in a single year
Solar power installed capacity (March 2026)	150.26 GW
Wind power installed capacity (March 2026)	56.09 GW
Global rank in renewable energy installed capacity	3rd (IRENA Renewable Energy Statistics 2026)
Highest-ever renewable share of electricity demand	51.5% (achieved on 29 July 2025)
India's 2030 non-fossil capacity target (NDC)	500 GW

(Figures sourced from the Ministry of New and Renewable Energy (MNRE), Press Information Bureau releases, and IRENA Renewable Energy Statistics 2026; official portals should be checked for the latest updates.)

Best Practices

- Switch to LED bulbs and BEE star-rated appliances
- Switch off lights, fans, and electronics when not in use
- Set air conditioners to an efficient temperature range (24–26°C)
- Service appliances regularly to maintain efficiency
- Explore rooftop solar eligibility under PM Surya Ghar
- Use natural daylight and ventilation wherever possible

Common Mistakes

- Leaving appliances on standby mode instead of switching them off completely
- Ignoring the BEE star rating when purchasing new appliances
- Overcooling or overheating rooms beyond comfortable, efficient levels
- Delaying appliance servicing, which reduces efficiency over time
- Assuming renewable energy adoption is only relevant at the industrial or government level

Myths vs Facts

Myth	Fact
"Renewable energy can't meet India's growing electricity demand"	Renewables already met over half of India's total electricity demand on a record day in July 2025, and capacity continues to expand rapidly
"Energy-efficient appliances are not worth the extra cost"	Higher-rated appliances typically pay back their price difference through electricity savings over their operating lifetime
"Rooftop solar is only viable for large buildings or businesses"	Government schemes like PM Surya Ghar specifically target individual households for rooftop solar adoption
"Switching off appliances at the plug makes no real difference"	Standby power draw across multiple devices in a home adds up to measurable, avoidable electricity consumption

Do's and Don'ts

Do	Don't
Switch off appliances when not in use	Don't leave devices running on standby unnecessarily
Choose BEE 5-star rated appliances	Don't ignore energy ratings while shopping
Service ACs and appliances regularly	Don't delay maintenance that affects efficiency
Explore rooftop solar eligibility	Don't assume renewables are only for large-scale users
Use natural light during the day	Don't rely on artificial lighting unnecessarily

Practical Tips

For Households

Switch to LED lighting throughout the home • Set ACs to 24–26°C • Unplug chargers and electronics not in active use

For Communities

Advocate for rooftop solar adoption in housing societies • Organise awareness sessions on PM Surya Ghar benefits • Encourage shared energy-efficient infrastructure like solar street lighting

For Schools & Offices

Install motion-sensor or timer-based lighting in common areas • Run periodic energy-audit awareness sessions • Choose energy-efficient equipment for new purchases

Practical Checklist

■	I have switched to LED bulbs at home
■	I check the BEE star rating before buying new appliances
■	I switch off lights, fans, and electronics when not in use
■	I know whether my home is eligible for rooftop solar subsidies
■	I service my air conditioner and major appliances regularly
■	I set my AC to an efficient temperature range
■	I have discussed energy-saving habits with family or peers

Quick Revision Notes

- Renewable energy = naturally replenishing sources (solar, wind, hydro); non-renewable = finite fossil fuels
- India's non-fossil capacity crossed 283 GW by March 2026, with solar as the leading source
- India met its 2030 non-fossil capacity milestone five years early, in June 2025
- BEE star ratings (1–5 stars) indicate appliance energy efficiency
- PM Surya Ghar subsidises rooftop solar for households; net metering credits surplus power
- MNRE, BEE, and the Ministry of Power are India's key energy regulatory bodies

Key Takeaways

1. India is undergoing one of the fastest clean energy transitions in the world, led by record solar and wind capacity additions

2. Conservation and renewable adoption are complementary, not competing, strategies for a sustainable energy future

3. Household-level choices — appliance ratings, usage habits, rooftop solar — genuinely add up at scale

4. Government schemes increasingly make renewable adoption accessible to individual households, not just large industry

5. Every unit of electricity saved through conservation reduces the total generation India needs to build

Glossary

- **BEE** — Bureau of Energy Efficiency, India's appliance energy-labelling authority
- **Distributed Renewable Energy (DRE)** — Small-scale renewable installations located close to where power is consumed
- **Grid** — The interconnected network transmitting electricity from source to consumer
- **Green Hydrogen** — Hydrogen fuel produced using renewable electricity, with no associated carbon emissions
- **Installed Capacity** — The maximum electricity output a power source can generate
- **Net Metering** — A billing system crediting rooftop solar owners for surplus electricity sent to the grid
- **Non-Fossil Capacity** — Combined renewable and nuclear power capacity
- **NDC** — Nationally Determined Contribution, a country's climate targets under the Paris Agreement
- **Photovoltaic (PV)** — Technology that converts sunlight directly into electricity
- **Star Rating** — India's official energy-efficiency label for appliances, from 1 to 5 stars

Frequently Asked Questions

1. What is the difference between renewable and non-renewable energy?
Renewable energy comes from naturally replenishing sources like sunlight, wind, and water; non-renewable energy comes from finite fossil fuels like coal, oil, and natural gas that take millions of years to form.

2. How much of India's power capacity is now non-fossil?
As of March 2026, India's non-fossil power capacity crossed 283 GW, more than half of the country's total installed electricity capacity.

3. What is PM Surya Ghar: Muft Bijli Yojana?
A national scheme launched in 2024 that subsidises rooftop solar installation, aiming to reach one crore households by FY 2026–27.

4. What does a BEE star rating mean?
It indicates an appliance's energy efficiency on a scale of 1 to 5 stars — the higher the rating, the less electricity the appliance consumes for the same output.

5. What is net metering?
A billing arrangement where a rooftop solar owner's meter credits them for any surplus electricity their system sends back to the grid.

6. Has India met its renewable energy targets?
India reached 50% of its cumulative electricity capacity from non-fossil sources in June 2025, five years ahead of its 2030 Paris Agreement target of 500 GW.

7. Which source contributes most to India's renewable capacity?
Solar power, at over 150 GW as of March 2026, is by far India's largest renewable energy source.

8. What is the National Green Hydrogen Mission?
A national mission launched in 2023 aiming to position India as a global hub for green hydrogen production and export.

9. Why does energy conservation matter even as renewables grow?
Every unit of electricity saved reduces the total generation capacity a country needs to build, making conservation the fastest and cheapest way to reduce both cost and emissions.

10. Which government body should I check for energy-saving appliance ratings?
The Bureau of Energy Efficiency (BEE), which administers India's official appliance star-rating system.

Official References

- Ministry of New and Renewable Energy (MNRE) — national renewable energy capacity data and programme information
- Bureau of Energy Efficiency (BEE) — appliance star-rating standards and energy conservation guidance
- Press Information Bureau (PIB) — releases on India's non-fossil capacity milestones
- International Renewable Energy Agency (IRENA) — Renewable Energy Statistics 2026
- Energy Conservation Act, 2001 (as amended 2022), and Electricity Act, 2003 — India Code portal

Official Government Website Links

- Ministry of New and Renewable Energy: **https://mnre.gov.in**
- PM Surya Ghar: Muft Bijli Yojana: **https://pmsuryaghar.gov.in**
- Bureau of Energy Efficiency: **https://beeindia.gov.in**
- Ministry of Power: **https://powermin.gov.in**
- Press Information Bureau: **https://pib.gov.in**

Conclusion

India's energy transition is one of the great infrastructure stories of this decade — but its success will ultimately be measured not just in gigawatts installed, but in habits changed. Every switched-off light, every star-rated appliance, every household that adopts rooftop solar adds up to a cleaner, more resilient energy system for the whole country.

Continue your Environment & Civic Responsibility journey with JanGyan. Stay efficient, stay curious, and remember: the energy you save is as valuable as the energy you generate.

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